Assignment 2 Process Report

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1 Schedule

We worked iteratively rather than in a strict waterfall. The first step was to read the assignment specification carefully and inspect the dataset structure. After that, we moved through exploration, feature engineering, modeling, evaluation, and report writing in several short cycles.

2 Task Allocation

The work was divided by responsibility, but all three members contributed to the final result and reviewed each other's output.

3 Reflection

Overall, the cooperation was effective because we separated the assignment into clear stages and kept the intermediate results reproducible. A major challenge was the dataset size, which required us to iterate on a manageable subset before producing the final run. Another challenge was aligning model quality with the fairness requirement from the assignment. That forced us to think beyond pure accuracy and to check whether the model behaved differently across user groups.

The most useful part of the collaboration was that each stage created input for the next one: the EDA informed the features, the validation results informed the fairness audit, and the report was written directly from the metrics produced by the pipeline. This reduced guesswork and made the final report easier to justify. If we had more time, we would further compare more ranking-oriented models and expand the fairness analysis to additional user segments.

Table 1. Work schedule for Assignment 2.

Date	Activity
23 April 2026	Read the assignment PDF, inspected the CSV schema, and confirmed the submission format.
23 April 2026	Performed exploratory data analysis, checked missing values, target imbalance, and price behavior.
23 April 2026	Built the first feature engineering pipeline and trained the popularity baseline and gradient boosting model.
23 April 2026	Evaluated NDCG@5 on a search-level validation split, then added fairness auditing and mitigation.
23 April 2026	Wrote the report in LNCS format, generated figures, and validated the LaTeX build.

Table 2. Main contributions of the group members.

Member	Contribution
Nicholas Boidi	Implemented the data pipeline, feature engineering, model training code, validation split, fairness analysis, and Kaggle submission generation.
Pablos Tselioudis Garmendia	Led the report structure, related work section, interpretation of results, and the discussion of scalable deployment.
Filippo Donghi	Helped with exploratory analysis, reviewed plots and metrics, and contributed to the final editing and presentation of the report.